Assignment: Frequent Itemsets

Big Data Analytics

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1 Introduction

In this assignment, we are interested in authors that frequently publish together. This can be useful for various reasons, such as identifying potential research groups or collaborations.

2 Dataset

We will use the publications from the DBLP dataset[1]. To facilitate the assignment, multiple preprocessed datasets of various sizes are already available on Blackboard as txt-files. In these files, each row represents the authors of a single publication, separated by a comma.

3 Assignment

The assignment consists of two parts. The first part focusses on identifying authors that frequently publish together using the A-Priori algorithm. The second part uses the results from the first part to identify association rules between authors.

3.1 Maximal author sets based on A-Priori

For a set of authors, the *support* is the number of publications in which all authors appear together. Next, let k be an integer with $k \geq 1$. A set S of k different authors is said to be a *maximal author set of size* k if there is no set S' of k authors with a support larger than the support of S. Stated differently, the maximal author sets of size k are the sets of authors of size k that appear together in the largest number of publications[2]. Your task is to identify the maximal author sets of size k for each value of k.

Write a Python program implementing the A-Priori algorithm to solve this task. Since A-Priori considers author sets of increasing size in each round, your program should be able to compute the maximal author sets of size k for all values of k in one run of the program, until no more maximal author sets are found.

Notice that for larger sets of authors, the support of these maximal author sets will typically be lower. Choosing a lower threshold in your implementation will therefore allow you to identify larger groups, but this will be more challenging as the number of frequent authors and frequent pairs will increase drastically. Your report should briefly discuss the threshold(s) you considered (cf. Section 4.2).

One might notice that all provided datasets are still relatively small in terms of the total number of publications. If desirable, it is allowed to keep the whole dataset in memory.

3.2 Association rules

Starting from the frequent itemsets obtained in the previous part, your task is to identify association rules between authors. For each frequent itemset S identified in the previous part, first compute the confidence of all possible association rules $S \setminus \{a\} \Rightarrow \{a\}$ for each author $a \in S$. Next, extend your program to identify association rules with more than one author on the right-hand side. Experiment with different confidence thresholds (e.g., 0.5, 0.8 and 1.0) and discuss the results in your report. Note that, for larger datasets, the total number of association rules might be too large to process when the number of authors on the right hand side increases. In these cases, you can increase the support threshold

[1] `https://dblp.org/`

[2] Note that this definition of maximal author set is *not* related to the notion of maximal frequent itemsets!

to reduce the number of rules generated and/or restrict your attention to association rules with a limited number of authors on the right-hand side. For each dataset and each confidence threshold, specify the support threshold you used, indicate the number of association rules found and provide a few examples of association rules with a high confidence.

4 Requirements

4.1 Implementation

The implementation must be in Python 3. For the first part, further refinements of the A-Priori algorithm or alternative approaches (e.g., one of the other algorithms considered in the lecture) are allowed. In this case, discuss in your report what you did, why you did it, and why the resulting algorithm is still correct.

You are allowed to use built-in functionality to facilitate your implementation (see the tips below for examples). **However, using libraries/tools that implement (large parts of) the frequent itemset and association rule mining algorithms for you is not allowed!**

Finally, you should properly structure and document your code.

Tips As sets are mutable in Python, they cannot be used as keys in dictionaries. Therefore, use `frozenset` instead. To generate all subsets of a certain size, use `itertools.combinations`. If your code is slow, identify the bottleneck before trying to optimize your approach (the `timeit` and/or `cProfile` library can be useful to identify bottlenecks). Make sure to discuss these bottlenecks and how you solved them in your report!

4.2 Report

Write a report covering all of the elements discussed below.

Data exploration For each dataset, provide a brief overview of the data. How many publications are there in the dataset? How many unique authors are there? What is the average number of authors co-authoring a publication? ...

Implementation details Discuss your implementation. What are the main challenges you encountered? How did you solve them? ...

Clearly mention and motivate important design choices (e.g., the chosen data structure to count pairs) and the chosen values for important parameters (e.g., the considered thresholds). **For the tiny and medium datasets, you should be able to process the data with a support threshold of 5 or lower. For the larger datasets, a support threshold of 25 or lower is expected.** With these parameters, you should be able to process the data within a reasonable amount of time (e.g., around 15 minutes for the largest dataset).

Results For each dataset, indicate whether your A-Priori and association rule mining algorithms were able to process the dataset. If you were unable to process one or more of the datasets, discuss the bottleneck and why you were unable to solve it. For the A-Priori algorithm, furthermore mention the threshold used for this dataset. For each $k \geq 1$, provide (i) the support of the maximal author sets of size k; (ii) the number of maximal author sets of size k; and (iii) a maximal author set of size k (one arbitrary example suffices if there are multiple maximal author sets of size k). For the association rule mining, specify for each dataset the considered support and confidence thresholds. For each confidence threshold, indicate the number of association rules found and provide a few examples.

4.3 Submission

You should hand in your code together with the report on Blackboard. Submission of the provided input data and raw output data is *not* required (instead, your report should summarize these results, cf. Section 4.2). **Make sure to include the name and student number of all group members in your report.**